

AWS Loan Default Pipeline

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Introduction:

This project aims to study the factors involved in predicting an individual defaulting on their loan. The general indicators like salary and credit score used are useful indicators but they are not sufficient in determining as other factors like family size, kids, etc. also highly influence the ability of an individual.

The dataset I have chosen is hosted on Kaggle ([link](#)). This is a dataset on the applicants for loans with their details and a 'TARGET' column specifying if they defaulted on their loan. There are a total of 122 columns, but we do not need all of them for the purpose of our analysis. The data is raw and needs cleaning and additional column building to be suitable for analysis and application of machine learning. There are 307,511 rows in the dataset. That's a good size to train machine learning models and plot meaningful visualizations to spot trends and patterns. The column 'TARGET' specifies if the loan was defaulted with one (1) or repaid successfully with zero (0).

These are the columns, with descriptions, that interest us for the project:

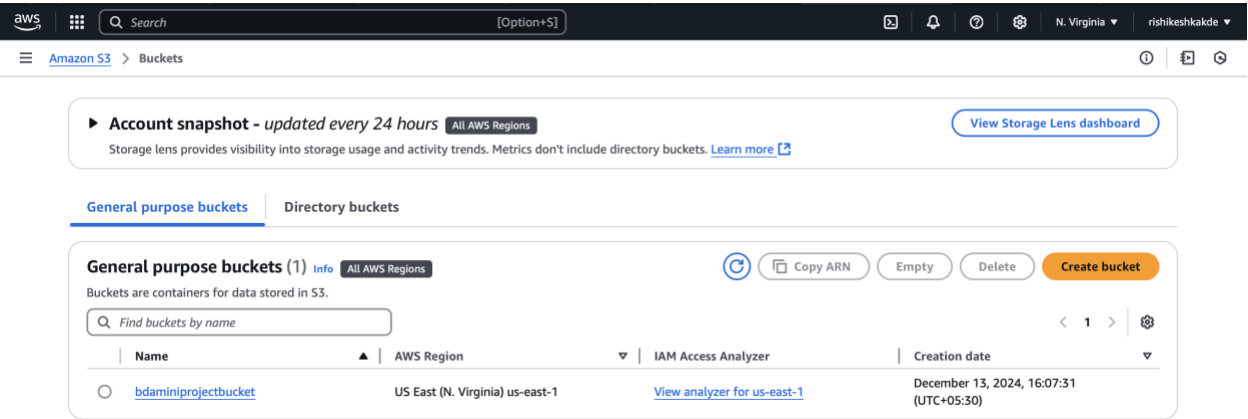
- NAME_HOUSING_TYPE: Type of housing the applicant resides in.
- AMT_CREDIT: Total credit amount of the loan.
- AMT_INCOME: Total income of the applicant.
- CNT_CHILDREN: Number of children the applicant has.
- CNT_FAM_MEMBERS: Total number of family members of the applicant.
- CODE_GENDER: Gender of the applicant.
- NAME_FAMILY_STATUS: Family status of the applicant.
- NAME_EDUCATION_TYPE: Education level of the applicant.
- NAME_INCOME_TYPE: Income type of the applicant.
- REGION_RATING_CLIENT: Regional rating of the client's area.
- OCCUPATION_TYPE: Occupation of the applicant.
- ORGANIZATION_TYPE: Type of organization the applicant works for.
- DAYS_BIRTH: Age of the applicant (in days).
- DAYS_EMPLOYED: Number of days the applicant has been employed.
- AMT_GOODS_PRICE: Price of the goods for which the loan is taken.

- AMT_INCOME_TOTAL: Total income of the applicant.

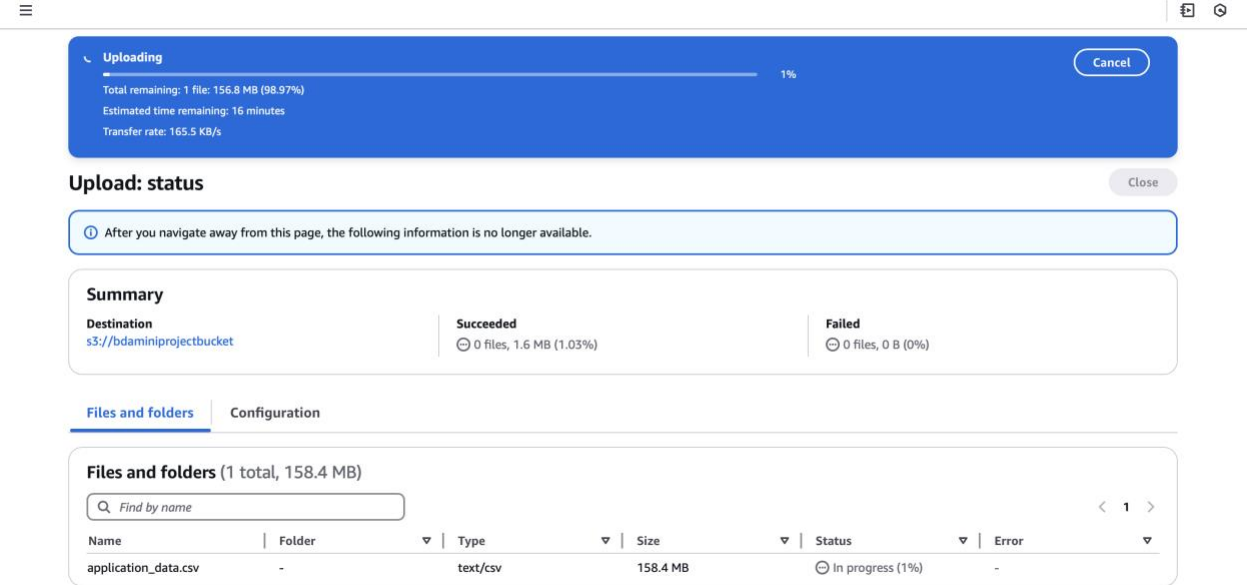
Methodology

I will use Amazon Web Services (AWS) technologies for this project, specifically, Amazon S3 to store the input and output files, EC2 for running the code for data cleaning, transformation and pre-processing in PySpark, Amazon SageMaker Studio to train ML model, and Amazon QuickSight to plot insightful visualization to make the case for useful predictors.

These are the steps I followed to setup the S3 and EC2 enviroment.
I Created an S3 Bucket called 'bdaminprojectbucket'



Uploading the file to S3 bucket:



S3 bucket after uploading the raw csv file:

aws

Search

[Option+S]

N. Virginia

rishikeshkakde

Upload succeeded

For more information, see the Files and folders table.

Close

Upload: status

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://badminiprojectbucket

Succeeded

1 file, 158.4 MB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 158.4 MB)

Find by name

Name	Folder	Type	Size	Status	Error
application_data.csv	-	text/csv	158.4 MB	Succeeded	-

Then, I Created an EC2 instance with Ubuntu AMI:

aws

Search

[Option+S]

Global

rishikeshkakde

EC2 Global View

Region explorer

Global search

Updated less than a minute ago

Global search (1/1)

Perform a global search to search for specific resources across all Regions for which your account is enabled

Download CSV

Manage

Find resources by attribute or tag

Resource Type = Instance

Clear filters

Name	Resource ID	Resource Type
BDA_MiniProject	i-00ea0b953eaeefb376	Instance

i-00ea0b953eaeefb376 (BDA_MiniProject) - eu-north-1

Details

Security groups

Volumes

Tags

Instance ID

i-00ea0b953eaeefb376

(BDA_MiniProject)

Public IPv4 address

16.171.38.0

Private IPv4 addresses

172.31.37.12

Public IPv4 DNS

ec2-16-171-38-0.eu-north-1.compute.amazonaws.com

open address

Instance type

t3.micro

VPC ID

vpc-06f1597d83faa64d0

Instance state

Running

Private IPv4 DNS

ip-172-31-37-12.eu-north-1.compute.internal

Launch Time

2024/12/11 17:44 GMT+5:30 (2 days)

Subnet ID

subnet-021a14c8c7800c330

Then I Connected to instance on the browser:

```
aws [Search] [Option+S] Stockholm rishikeshkakde

Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1020-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Fri Dec 13 11:24:26 UTC 2024

System load: 0.1          Temperature: -273.1 C
Usage of /:  14.4% of 28.02GB    Processes:   109
Memory usage: 25%              Users logged in: 0
Swap usage:  0%                IPv4 address for ens5: 172.31.37.12

* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

16 updates can be applied immediately.
9 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Dec 11 12:14:48 2024 from 13.48.4.203
ubuntu@ip-172-31-37-12:~$
```

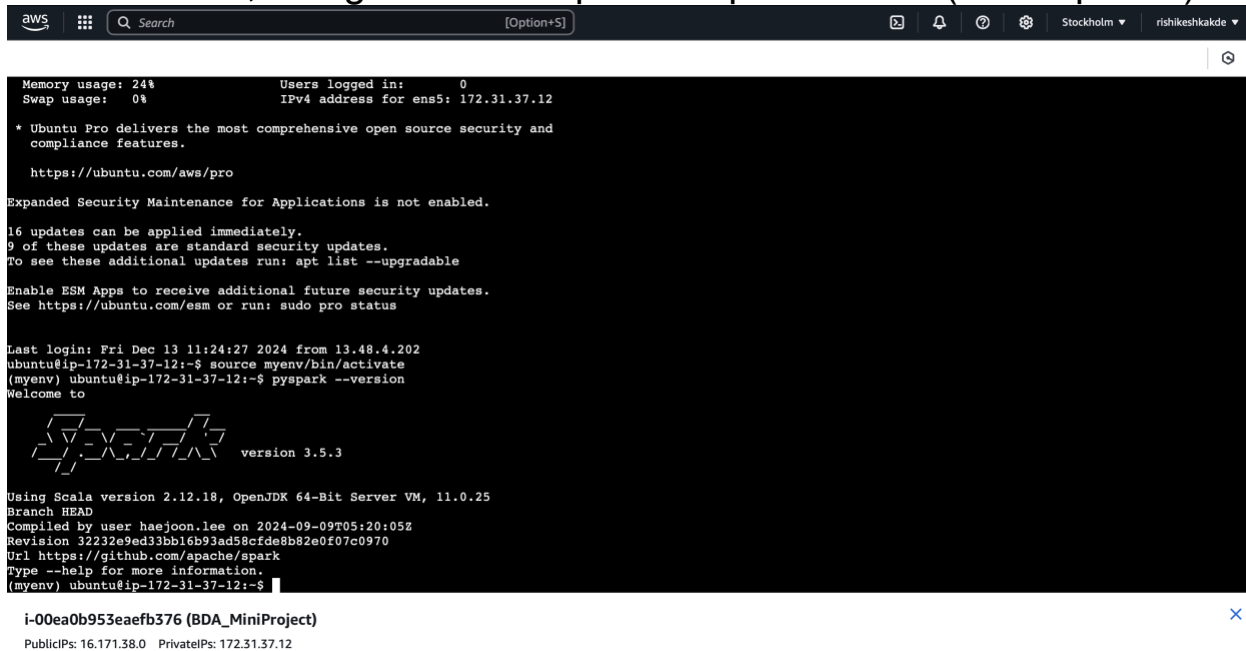
i-00ea0b953eae6b376 (BDA_MiniProject)

PublicIPs: 16.171.38.0 PrivateIPs: 172.31.37.12

I also connected to AWS using AWS CLI on my Mac with my credentials:

```
rishikeshkakde@Rishikeshs-MacBook-Pro-6 ~ % aws s3 ls
2024-12-13 16:07:32 bdaminiprojectbucket
rishikeshkakde@Rishikeshs-MacBook-Pro-6 ~ %
```

In the EC2 instance, I Installed Apache Pyspark in the virtual environment in the EC2 Instance, along with the required dependencies (Hadoop-aws):



```
Memory usage: 24%      Users logged in: 0
Swap usage:  0%       IPv4 address for ens5: 172.31.37.12

* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

  https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

16 updates can be applied immediately.
9 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Dec 13 11:24:27 2024 from 13.48.4.202
ubuntu@ip-172-31-37-12:~$ source myenv/bin/activate
(myenv) ubuntu@ip-172-31-37-12:~$ pyspark --version
Welcome to

      ____              __
     /  _/   ____  ____/  /
    /  /_  / __ \ / __ \_/_/
   /___/_/ /_/ / /_/ /___/
          /_/

version 3.5.3

Using Scala version 2.12.18, OpenJDK 64-Bit Server VM, 11.0.25
Branch HEAD
Compiled by user haejoon.lee on 2024-09-09T05:20:05Z
Revision 32232e9ed33bb16b93ad58cfd8b82e0f07c0970
Url https://github.com/apache/spark
Type --help for more information.
(myenv) ubuntu@ip-172-31-37-12:~$
```

i-00ea0b953eae6b376 (BDA_MiniProject)

PublicIPs: 16.171.38.0 PrivateIPs: 172.31.37.12

I installed aws-cli on my local machine and in the EC2 instance too (in a python virtual environment where I have also installed PySpark).

I have put a screenshot before of it configured on my local. I used my local machine to SSH into my EC2 instance too. Here is a screenshot of the EC2 instance from my local machine (Macbook):

```

Last login: Fri Dec 13 23:15:02 on ttys002
[rishtkeshkakde@Rishikeshs-MacBook-Pro-6 Mini-Project % ssh -i MiniProject_KP.pem ubuntu@16.171.38.0

The authenticity of host '16.171.38.0 (16.171.38.0)' can't be established.
ED25519 key fingerprint is SHA256:pcedggner4sFGq1/tsRKuFxc1aJw1KpSYGS2IOIQiw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '16.171.38.0' (ED25519) to the list of known hosts.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@      WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'MiniProject_KP.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "MiniProject_KP.pem": bad permissions
ubuntu@16.171.38.0: Permission denied (publickey).
rishtkeshkakde@Rishikeshs-MacBook-Pro-6 Mini-Project % chmod 400 MiniProject_KP.pem
[rishtkeshkakde@Rishikeshs-MacBook-Pro-6 Mini-Project % ssh -i MiniProject_KP.pem ubuntu@16.171.38.0
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1020-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Dec 13 18:21:45 UTC 2024

System load:  0.0           Temperature:   -273.1 C
Usage of /:   14.7% of 28.0GB Processes:      111
Memory usage: 30%          Users logged in: 1
Swap usage:   0%           IPv4 address for ens5: 172.31.37.12

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

16 updates can be applied immediately.
9 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Dec 13 18:11:11 2024 from 13.48.4.202
ubuntu@ip-172-31-37-12:~$

```

After logging into my EC2 instance using my Mac's terminal, I sourced the python virtual environment and launched PySpark with the following command:

```
pyspark --packages org.apache.hadoop:hadoop-aws:3.3.1
```

The challenge I faced here was that I was not able to pull data from the S3 bucket because my environment variable was not set. I set them using the following exports:

```
export AWS_ACCESS_KEY_ID="AKIAUJ3VT6YGT3XZX3EZ"
```

export

AWS_SECRET_ACCESS_KEY="10Uk/dFFQ0VdEtDbjTwmeRU3ly6JxtvwIzkMauD5"

After this, even in the PySpark environment I was able to get data from the S3 bucket.

Here is a screenshot of successful load of data from S3 bucket in PySpark and display of first five column's first 5 rows.

```
(myenv) ubuntu@ip-172-31-37-12:~$ pyspark --packages org.apache.hadoop:hadoop-aws:3.3.1
Python 3.12.3 (main, Nov 6 2024, 18:32:19) [GCC 13.2.0] on linux
Type "help", "copyright", "credits" or "license()" for more information.
:: loading settings :: url = jar:file:/home/ubuntu/myenv/lib/python3.12/site-packages/pyspark/jars/ivy-2.5.1.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /home/ubuntu/.ivy2/cache
The jars for the packages stored in: /home/ubuntu/.ivy2/jars
org.apache.hadoop:hadoop-aws added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-c3926a2f-989e-4ef7-b537-9ca8a2ba71de;1.0
  confs: [default]
    found org.apache.hadoop:hadoop-aws:3.3.1 in central
    found com.amazonaws:aws-java-sdk-bundle:1.11.901 in central
    found org.wildfly.openssl:wildfly-openssl:1.0.7.Final in central
:: resolution report :: resolve 340ms :: artifacts dl 13ms
  :: modules in use:
    com.amazonaws:aws-java-sdk-bundle:1.11.901 from central in [default]
    org.apache.hadoop:hadoop-aws:3.3.1 from central in [default]
    org.wildfly.openssl:wildfly-openssl:1.0.7.Final from central in [default]
-----
| conf | number | search | dwnlded | evicted | number | dwnlded |
-----+-----+-----+-----+-----+-----+-----+
| default | 3 | 0 | 0 | 0 | 3 | 0 |
-----
:: retrieving :: org.apache.spark#spark-submit-parent-c3926a2f-989e-4ef7-b537-9ca8a2ba71de
  confs: [default]
  0 artifacts copied, 3 already retrieved (0kB/5ms)
24/12/13 22:29:40 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to

      ____
     /   _ \
    /   /___ \
   /   /_  __ \
  /   /___/ _ \
 /   /___/ ___ \
/   /___/_/___ \
/___/_/___/_/___ \
version 3.5.3

Using Python version 3.12.3 (main, Nov 6 2024 18:32:19)
Spark context Web UI available at http://ip-172-31-37-12.eu-north-1.compute.internal:4040
Spark context available as 'sc' (master = local[*], app id = local-1734128984514).
SparkSession available as 'spark'.
>>> df = spark.read.csv("s3a://bda-miniprojectbucket/application_data.csv", header=True, inferSchema=True)
24/12/13 22:29:57 WARN MetricsConfig: Cannot locate configuration: tried hadoop-metrics2-s3a-file-system.properties,hadoop-metrics2.properties
>>> # Select the first 5 columns
>>> columns_to_display = df.columns[:5]
>>> df_selected = df.select(*columns_to_display)
>>> # Show the first 5 rows
>>> df_selected.show(5)
+-----+-----+-----+-----+-----+
|SK_ID_CURR|TARGET|NAME_CONTRACT_TYPE|CODE_GENDER|FLAG_OWN_CAR|
+-----+-----+-----+-----+-----+
| 100002 | 1 | Cash loans | M | N |
| 100003 | 0 | Cash loans | F | N |
| 100004 | 0 | Revolving loans | M | Y |
| 100006 | 0 | Cash loans | F | N |
| 100007 | 0 | Cash loans | M | N |
+-----+-----+-----+-----+-----+
only showing top 5 rows

>>> █
```

To clean the dataset and create two new columns – ‘AGE’, ‘AGE_GROUP’ and ‘AMT_CREDIT_RANGE’, ‘AMT_INCOME_RANGE’, ‘YEARS_EMPLOYED’, and ‘EMPLOYMENT_YEAR’.

I made a PySpark file ‘DataCleaning_Transformation_Aggregation.py’. Please find the file along with this submission this is what the script does.

It performs data cleaning and transformation to prepare a dataset for analysis and modeling. Negative values in date-related columns are converted to absolute values, ensuring consistency. Continuous numerical variables, such as income, credit, age, and employment duration, are transformed into categorical bins for better interpretability. Missing data is handled by filling null values in categorical columns with the mode or a placeholder like "Unknown," and numerical columns with their median values. The script then selects a subset of relevant, cleaned columns for analysis and saves the transformed dataset as csv in the EC2 instance. After cleaning, it calculates the following five key metrics and saves the five results as csv files in the instance. I tried saving everything back to S3 bucket but it was taking too long. So manually upload the files to S3 bucket after they are generated.

Metric Descriptions and CSV File Names:

1. Metric 1: Top 3 Longest Employment Durations
 - a. Description: Extracts the top 3 applicants with the longest employment durations based on the DAYS_EMPLOYED column.
 - b. CSV File Name: AggregatedMetric-1.csv
2. Metric 2: Average Income by Education Type
 - a. Description: Calculates the average total income (AMT_INCOME_TOTAL) grouped by the NAME_EDUCATION_TYPE column.
 - b. CSV File Name: AggregatedMetric-2.csv
3. Metric 3: Number of Applicants by Family Status
 - a. Description: Counts the number of applicants for each category in the NAME_FAMILY_STATUS column.
 - b. CSV File Name: AggregatedMetric-3.csv
4. Metric 4: Average Credit Amount by Housing Type
 - a. Description: Calculates the average credit amount (AMT_CREDIT) grouped by the NAME_HOUSING_TYPE column.
 - b. CSV File Name: AggregatedMetric-4.csv
5. Metric 5: Top 3 Occupations with the Most Applicants
 - a. Description: Identifies the top 3 occupations with the highest number of applicants based on the OCCUPATION_TYPE column.
 - b. CSV File Name: AggregatedMetric-5.csv

I wrote the code on my local machine, then used AWS-CLI to login to the EC2 instance from my local terminal and then used 'vim' to create a file 'DataCleaning_Transformation_Aggregation.py' in the instance. Then simply copy pasted the code from my VS Code on local to the EC2 instance's newly created file.

After that I ran the code to generate the cleaned file and the five key metrics files.

Here's a screenshot of the successful run's end log and generated files:

```
AggregatedMetric-5 calculated and saved as 'AggregatedMetric-5.csv'.
24/12/14 08:02:16 INFO SparkContext: SparkContext is stopping with exitCode 0.
24/12/14 08:02:16 INFO SparkUI: Stopped Spark web UI at http://ip-172-31-37-12.eu-north-1.compute.internal:4040
24/12/14 08:02:17 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
24/12/14 08:02:17 INFO MemoryStore: MemoryStore cleared
24/12/14 08:02:17 INFO BlockManager: BlockManager stopped
24/12/14 08:02:17 INFO BlockManagerMaster: BlockManagerMaster stopped
24/12/14 08:02:17 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
24/12/14 08:02:17 INFO SparkContext: Successfully stopped SparkContext
Spark session stopped.
24/12/14 08:02:17 INFO ShutdownHookManager: Shutdown hook called
24/12/14 08:02:17 INFO ShutdownHookManager: Deleting directory /tmp/spark-55e0319f-e24c-4294-b2fc-5b70195d4754/pyspark-bef670a4-d262-42e8-ab32-de670a1e7a7b
24/12/14 08:02:17 INFO ShutdownHookManager: Deleting directory /tmp/spark-55e0319f-e24c-4294-b2fc-5b70195d4754
24/12/14 08:02:17 INFO ShutdownHookManager: Deleting directory /tmp/spark-b19c49f7-4cd9-43e2-bf44-3ed965921b4d
24/12/14 08:02:17 INFO MetricsSystemImpl: Stopping s3a-file-system metrics system...
24/12/14 08:02:17 INFO MetricsSystemImpl: s3a-file-system metrics system stopped.
24/12/14 08:02:17 INFO MetricsSystemImpl: s3a-file-system metrics system shutdown complete.
(myenv) ubuntu@ip-172-31-37-12:~$ ls
AggregatedMetric-1.csv AggregatedMetric-2.csv AggregatedMetric-3.csv AggregatedMetric-4.csv AggregatedMetric-5.csv DataCleaning_Transformation_Aggregation.py cleaned_application_data.csv myenv
(myenv) ubuntu@ip-172-31-37-12:~$
```

Here is how I moved these files to S3 bucket:

1. I created a zip of all the relevant files on the EC2 instance:

```
(myenv) ubuntu@ip-172-31-37-12:~$ ls
AggregatedMetric-1.csv AggregatedMetric-2.csv AggregatedMetric-3.csv AggregatedMetric-4.csv AggregatedMetric-5.csv DataCleaning_Transformation_Aggregation.py cleaned_application_data.csv myenv
(myenv) ubuntu@ip-172-31-37-12:~$ zip -r project_files.zip AggregatedMetric-*.csv DataCleaning_Transformation_Aggregation.py cleaned_application_data.csv
adding: AggregatedMetric-1.csv/ (stored 0%)
adding: AggregatedMetric-1.csv/part-00000-984e4ff6-ce6e-45a5-b131-3a0c90298901-c000.csv.crc (stored 0%)
adding: AggregatedMetric-1.csv/_SUCCESS.crc (stored 0%)
adding: AggregatedMetric-1.csv/_SUCCESS (stored 0%)
adding: AggregatedMetric-1.csv/part-00000-984e4ff6-ce6e-45a5-b131-3a0c90298901-c000.csv (deflated 34%)
adding: AggregatedMetric-2.csv/ (stored 0%)
adding: AggregatedMetric-2.csv/_SUCCESS.crc (stored 0%)
adding: AggregatedMetric-2.csv/_SUCCESS (stored 0%)
adding: AggregatedMetric-2.csv/part-00000-6cd3d587-963e-4e4c-a5ea-087f3a28d773-c000.csv.crc (stored 0%)
adding: AggregatedMetric-2.csv/part-00000-6cd3d587-963e-4e4c-a5ea-087f3a28d773-c000.csv (deflated 21%)
adding: AggregatedMetric-3.csv/ (stored 0%)
adding: AggregatedMetric-3.csv/part-00000-a2855bf1-bfb2-4518-bf3a-62a166fa7d1e-c000.csv.crc (stored 0%)
adding: AggregatedMetric-3.csv/_SUCCESS.crc (stored 0%)
adding: AggregatedMetric-3.csv/_SUCCESS (stored 0%)
adding: AggregatedMetric-3.csv/part-00000-a2855bf1-bfb2-4518-bf3a-62a166fa7d1e-c000.csv (deflated 9%)
adding: AggregatedMetric-4.csv/ (stored 0%)
adding: AggregatedMetric-4.csv/_SUCCESS.crc (stored 0%)
adding: AggregatedMetric-4.csv/_SUCCESS (stored 0%)
adding: AggregatedMetric-4.csv/part-00000-145ba712-fec0-4382-beae-60797719211b-c000.csv.crc (stored 0%)
adding: AggregatedMetric-4.csv/part-00000-145ba712-fec0-4382-beae-60797719211b-c000.csv (deflated 26%)
adding: AggregatedMetric-5.csv/ (stored 0%)
adding: AggregatedMetric-5.csv/_SUCCESS.crc (stored 0%)
adding: AggregatedMetric-5.csv/part-00000-d021972c-5195-443f-830a-8e9fe8f202f5-c000.csv (stored 0%)
adding: AggregatedMetric-5.csv/_SUCCESS (stored 0%)
adding: AggregatedMetric-5.csv/part-00000-d021972c-5195-443f-830a-8e9fe8f202f5-c000.csv.crc (stored 0%)
adding: DataCleaning_Transformation_Aggregation.py (deflated 69%)
adding: cleaned_application_data.csv/ (stored 0%)
adding: cleaned_application_data.csv/_SUCCESS.crc (stored 0%)
adding: cleaned_application_data.csv/part-00000-74fd085a-afc9-47ae-a36d-31f53a33029d-c000.csv.crc (deflated 0%)
adding: cleaned_application_data.csv/_SUCCESS (stored 0%)
adding: cleaned_application_data.csv/part-00000-74fd085a-afc9-47ae-a36d-31f53a33029d-c000.csv (deflated 78%)
(myenv) ubuntu@ip-172-31-37-12:~$ ls
AggregatedMetric-1.csv AggregatedMetric-3.csv AggregatedMetric-5.csv cleaned_application_data.csv project_files.zip
AggregatedMetric-2.csv AggregatedMetric-4.csv DataCleaning_Transformation_Aggregation.py myenv
(myenv) ubuntu@ip-172-31-37-12:~$
```

2. Then I downloaded the zip file on my local like this (to have copies to submission):

```
rishikeshkakke@Mac-462 ~ % cd Desktop
rishikeshkakke@Mac-462 Desktop % scp -i MiniProject_KP.pem ubuntu@16.171.38.0:/home/ubuntu/project_files.zip .
project_files.zip 100% 1249KB 764.6KB/s 00:01
rishikeshkakke@Mac-462 Desktop %
```

3. I Realized that these files are folders with files ‘_SUCCESS’ and a randomly assigned csv name in them, with the csv file being the one that I want. I renamed the csv files in all the folders with appropriate names and uploaded them manually to S3 bucket.

Name	Date Modified	Size	Kind
_SUCCESS	Today at 1:31 PM	Zero bytes	Document
part-00000-984e4ff6...0c90298901-c000.csv	Today at 1:31 PM	35 bytes	CSV Document

EC2 Files			
Name	Date Modified	Size	Kind
AggregatedMetric-1.csv	Today at 1:31 PM	35 bytes	CSV Document
AggregatedMetric-2.csv	Today at 1:32 PM	215 bytes	CSV Document
AggregatedMetric-3.csv	Today at 1:32 PM	117 bytes	CSV Document
AggregatedMetric-4.csv	Today at 1:32 PM	235 bytes	CSV Document
AggregatedMetric-5.csv	Today at 1:32 PM	76 bytes	CSV Document
cleaned_application_data.csv	Today at 1:31 PM	5.7 MB	CSV Document

4. I then uploaded all the respective files to the S3 bucket:

Upload succeeded
For more information, see the Files and folders table.

Summary

Destination

s3://bdaminiprojectbucket

Succeeded

6 files, 48.4 MB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

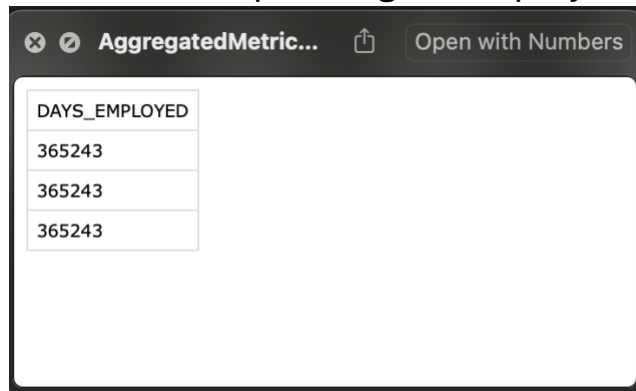
Files and folders (6 total, 48.4 MB)

Find by name

Name	Folder	Type	Size	Status	Error
AggregatedMetric-2.csv	-	text/csv	224.0 B	Succeeded	-
AggregatedMetric-3.csv	-	text/csv	136.0 B	Succeeded	-
AggregatedMetric-1.csv	-	text/csv	35.0 B	Succeeded	-
AggregatedMetric-4.csv	-	text/csv	240.0 B	Succeeded	-
AggregatedMetric-5.csv	-	text/csv	79.0 B	Succeeded	-
cleaned_application_data.c...	-	text/csv	48.4 MB	Succeeded	-

All the files can be found with this submission. Here are the screenshots for the aggregated metrics:

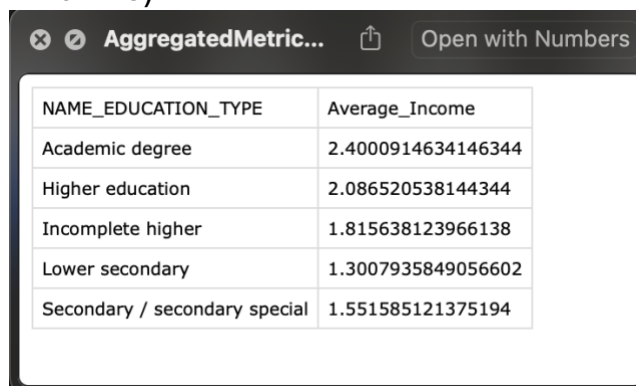
- Metric 1: Top 3 Longest Employment Durations



A screenshot of a Numbers spreadsheet window titled 'AggregatedMetric...'. The spreadsheet displays a single column with the header 'DAYS_EMPLOYED' and three rows, each containing the value '365243'.

DAYS_EMPLOYED
365243
365243
365243

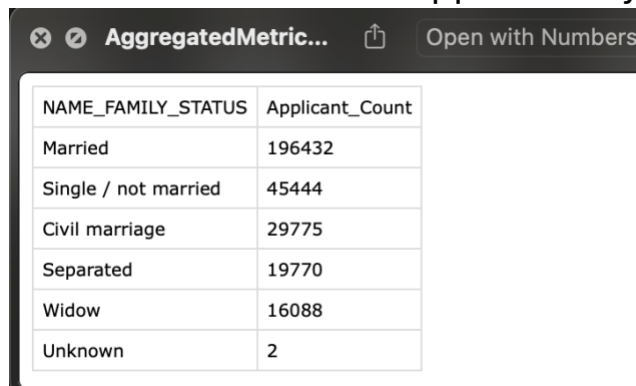
- Metric 2: Average Income by Education Type (Average_ Income Column in 100k units)



A screenshot of a Numbers spreadsheet window titled 'AggregatedMetric...'. The spreadsheet displays a table with two columns: 'NAME_EDUCATION_TYPE' and 'Average_Income'. The data is as follows:

NAME_EDUCATION_TYPE	Average_Income
Academic degree	2.4000914634146344
Higher education	2.086520538144344
Incomplete higher	1.815638123966138
Lower secondary	1.3007935849056602
Secondary / secondary special	1.551585121375194

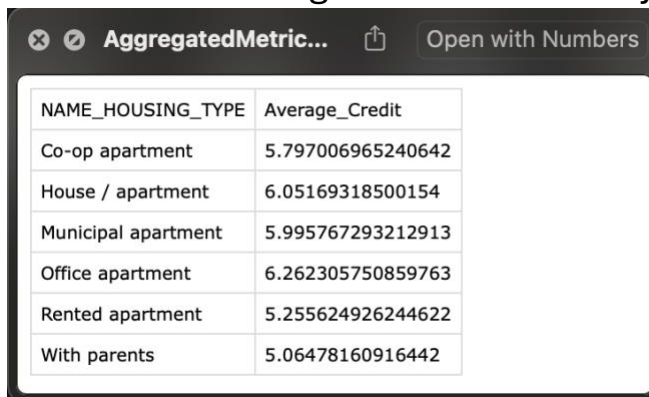
- Metric 3: Number of Applicants by Family Status



A screenshot of a Numbers spreadsheet window titled 'AggregatedMetric...'. The spreadsheet displays a table with two columns: 'NAME_FAMILY_STATUS' and 'Applicant_Count'. The data is as follows:

NAME_FAMILY_STATUS	Applicant_Count
Married	196432
Single / not married	45444
Civil marriage	29775
Separated	19770
Widow	16088
Unknown	2

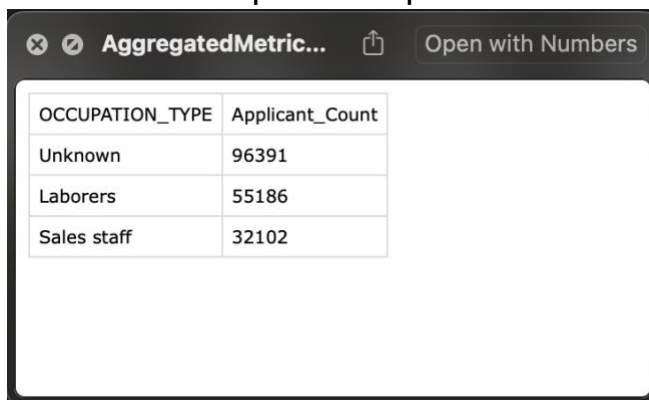
- Metric 4: Average Credit Amount by Housing Type



The screenshot shows a Numbers spreadsheet window titled 'AggregatedMetric...'. It contains a table with two columns: 'NAME_HOUSING_TYPE' and 'Average_Credit'. The table lists seven housing types and their corresponding average credit amounts.

NAME_HOUSING_TYPE	Average_Credit
Co-op apartment	5.797006965240642
House / apartment	6.05169318500154
Municipal apartment	5.995767293212913
Office apartment	6.262305750859763
Rented apartment	5.255624926244622
With parents	5.06478160916442

- Metric 5: Top 3 Occupations with the Most Applicants



The screenshot shows a Numbers spreadsheet window titled 'AggregatedMetric...'. It contains a table with two columns: 'OCCUPATION_TYPE' and 'Applicant_Count'. The table lists the top three occupations by applicant count.

OCCUPATION_TYPE	Applicant_Count
Unknown	96391
Laborers	55186
Sales staff	32102

Now, I created a new PySpark script with the name 'SQL_Analysis_PySpark.py' that runs queries to derive the following insights from the cleaned dataset.

1. Top Regions by Client Ratings: Understand which regions have the most highly rated clients.
2. Income Trends Across Demographics: Analyze income distribution across different NAME_EDUCATION_TYPE or NAME_INCOME_TYPE.
3. Popular Occupation Types: Determine the most common occupations among applicants.
4. Credit Amount Patterns by Housing Type: Explore the average credit amount taken by applicants based on their NAME_HOUSING_TYPE.
5. Loan Trends Across Age Groups: Loan distribution among different age groups.

Please find the script for this along with this submission.

I ran the script with the following command on the instance.

```
spark-submit --packages org.apache.hadoop:hadoop-aws:3.3.1,com.amazonaws:aws-java-sdk-bundle:1.11.901 SQL_Analysis_PySpark.py
```

Screenshot of the script kickoff:

```
(myenv) ubuntu@ip-172-31-37-12:~$ spark-submit --packages org.apache.hadoop:hadoop-aws:3.3.1,com.amazonaws:aws-java-sdk-bundle:1.11.901 SQL_Analysis_PySpark.py
:: loading settings :: url = jar:file:/home/ubuntu/myenv/lib/python3.12/site-packages/pyspark/jars/ivy-2.5.1.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /home/ubuntu/.ivy2/cache
The jars for the packages stored in: /home/ubuntu/.ivy2/jars
org.apache.hadoop:hadoop-aws added as a dependency
com.amazonaws:aws-java-sdk-bundle added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-b92b994e-c7e0-4fd9-bb29-397b6f2c4eaf;1.0
  confs: [default]
    found org.apache.hadoop:hadoop-aws;3.3.1 in central
    found com.amazonaws:aws-java-sdk-bundle;1.11.901 in central
    found org.wildfly.openssl#wildfly-openssl;1.0.7.Final in central
:: resolution report :: resolve 464ms :: artifacts d1 19ms
:: modules in use:
  com.amazonaws:aws-java-sdk-bundle;1.11.901 from central in [default]
  org.apache.hadoop:hadoop-aws;3.3.1 from central in [default]
  org.wildfly.openssl#wildfly-openssl;1.0.7.Final from central in [default]
-----
| conf | number | search | downlded | evicted | artifacts | number | downlded |
-----+-----+-----+-----+-----+-----+-----+-----
| default | 3 | 0 | 0 | 0 | 3 | 0 |
-----
:: retrieving :: org.apache.spark#spark-submit-parent-b92b994e-c7e0-4fd9-bb29-397b6f2c4eaf
  confs: [default]
```

Screenshot of script completion:

```
24/12/14 11:13:35 INFO FileFormatWriter: Write job af53f44e-da39-4399-8bae-feb5e151eb00 committed. Elapsed time: 10 ms.
24/12/14 11:13:35 INFO FileFormatWriter: Finished processing stats for write job af53f44e-da39-4399-8bae-feb5e151eb00.
Query 5 completed. Results saved to 'LoanTrendsByAgeGroup.csv'.
24/12/14 11:13:35 INFO SparkContext: SparkContext is stopping with exitCode 0.
24/12/14 11:13:35 INFO SparkUI: Stopped Spark web UI at http://ip-172-31-37-12.eu-north-1.compute.internal:4040
24/12/14 11:13:35 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
24/12/14 11:13:35 INFO MemoryStore: MemoryStore cleared
24/12/14 11:13:35 INFO BlockManager: BlockManager stopped
24/12/14 11:13:35 INFO BlockManagerMaster: BlockManagerMaster stopped
24/12/14 11:13:35 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
24/12/14 11:13:35 INFO SparkContext: Successfully stopped SparkContext
Data analysis completed and Spark session stopped.
24/12/14 11:13:35 INFO ShutdownHookManager: Shutdown hook called
24/12/14 11:13:35 INFO ShutdownHookManager: Deleting directory /tmp/spark-bfca9f26-d12a-40f5-a097-10e605bec2b7
24/12/14 11:13:35 INFO ShutdownHookManager: Deleting directory /tmp/spark-317c3d85-07e6-4034-8c2a-2f99a3c6435d/pyspark-e8031605-12e7-44c7-985d-0e29782b3187
24/12/14 11:13:35 INFO ShutdownHookManager: Deleting directory /tmp/spark-317c3d85-07e6-4034-8c2a-2f99a3c6435d
24/12/14 11:13:35 INFO MetricsSystemImpl: Stopping s3a-file-system metrics system...
24/12/14 11:13:35 INFO MetricsSystemImpl: s3a-file-system metrics system stopped.
24/12/14 11:13:35 INFO MetricsSystemImpl: s3a-file-system metrics system shutdown complete.
(myenv) ubuntu@ip-172-31-37-12:~$ ls
CreditPatternsByHousing.csv      IncomeTrendsByEducation.csv      PopularOccupationTypes.csv      TopRegionsByClientRatings.csv
DataCleaning_Transformation_Aggregation.py  LoanTrendsByAgeGroup.csv         SQL_Analysis_PySpark.py         myenv
(myenv) ubuntu@ip-172-31-37-12:~$
```

I zipped these files to a folder and downloaded on my local system and then uploaded to S3 bucket from using browser upload:

Zipping the files on EC2 Instance:

```
(myenv) ubuntu@ip-172-31-37-12:~$ zip my_project_files.zip CreditPatternsByHousing.csv IncomeTrendsByEducation.csv PopularOccupationTypes.csv TopRegionsByClientRatings.csv LoanTrendsByAgeGroup.csv
adding: CreditPatternsByHousing.csv/ (stored 0%)
adding: IncomeTrendsByEducation.csv/ (stored 0%)
adding: PopularOccupationTypes.csv/ (stored 0%)
adding: TopRegionsByClientRatings.csv/ (stored 0%)
adding: LoanTrendsByAgeGroup.csv/ (stored 0%)
(myenv) ubuntu@ip-172-31-37-12:~$
```

Downloading it to local:

```
rishikeshkakde — zsh — 112x24

rishikeshkakde@Mac-491 ~ % scp -i Desktop/MiniProject_KP.pem ubuntu@16.171.38.0:~/my_project_files.zip Desktop/
my_project_files.zip
100% 938 2.5KB/s 00:00

rishikeshkakde@Mac-491 ~ %
```

Also uploading the Spark SQL Results file to the S3 bucket:

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and various utility icons. Below the navigation bar, a green banner indicates "Upload succeeded" for the destination `s3://bdaminprojectbucket`. A blue banner below it states: "After you navigate away from this page, the following information is no longer available." The main content area has a "Summary" section showing 5 files (807.0 B) succeeded and 0 files (0 B) failed. Below this, the "Files and folders" tab is selected, displaying a table of 5 files uploaded to the "Spark SQL Outputs/" folder.

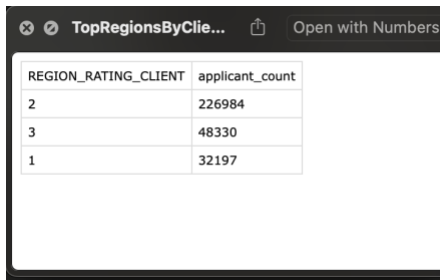
Name	Folder	Type	Size	Status	Error
PopularOccupationTypes.csv	Spark SQL Outputs/	text/csv	111.0 B	Succeeded	-
IncomeTrendsByEducation...	Spark SQL Outputs/	text/csv	221.0 B	Succeeded	-
LoanTrendsByAgeGroup.csv	Spark SQL Outputs/	text/csv	173.0 B	Succeeded	-
TopRegionsByClientRatings...	Spark SQL Outputs/	text/csv	62.0 B	Succeeded	-
CreditPatternsByHousing.csv	Spark SQL Outputs/	text/csv	240.0 B	Succeeded	-

Here are the results and queries that were in the PySpark Notebook:

Query 1: Top Regions by Client Ratings

```
SELECT REGION_RATING_CLIENT, COUNT(*) AS applicant_count
FROM application_data
GROUP BY REGION_RATING_CLIENT
ORDER BY applicant_count DESC
```

Result 1: 'TopRegionsByClientRatings.csv'

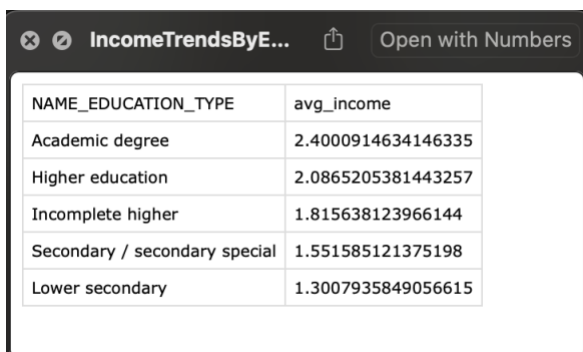


REGION_RATING_CLIENT	applicant_count
2	226984
3	48330
1	32197

Query 2: Income Trends Across Education Types

```
SELECT NAME_EDUCATION_TYPE, AVG(AMT_INCOME_TOTAL) AS avg_income
FROM application_data
GROUP BY NAME_EDUCATION_TYPE
ORDER BY avg_income DESC
```

Result 2: 'IncomeTrendsByEducation.csv'



NAME_EDUCATION_TYPE	avg_income
Academic degree	2.4000914634146335
Higher education	2.0865205381443257
Incomplete higher	1.815638123966144
Secondary / secondary special	1.551585121375198
Lower secondary	1.3007935849056615

Query 3: Popular Occupation Types

```
SELECT OCCUPATION_TYPE, COUNT(*) AS applicant_count
FROM application_data
GROUP BY OCCUPATION_TYPE
ORDER BY applicant_count DESC
LIMIT 5
```

Result 3: 'PopularOccupationTypes.csv'

OCCUPATION_TYPE	applicant_count
Unknown	96391
Laborers	55186
Sales staff	32102
Core staff	27570
Managers	21371

Query 4: Credit Amount Patterns by Housing Type

```
SELECT NAME_HOUSING_TYPE, AVG(AMT_CREDIT) AS avg_credit
FROM application_data
GROUP BY NAME_HOUSING_TYPE
ORDER BY avg_credit DESC
```

Result 4: 'CreditPatternsByHousing.csv'

NAME_HOUSING_TYPE	avg_credit
Office apartment	6.262305750859753
House / apartment	6.0516931850009374
Municipal apartment	5.995767293212909
Co-op apartment	5.797006965240638
Rented apartment	5.2556249262446135
With parents	5.064781609164415

Query 5: Loan Trends Across Age Groups

```
SELECT AGE_GROUP, AVG(AMT_CREDIT) AS avg_credit, COUNT(*) AS
applicant_count
FROM application_data
GROUP BY AGE_GROUP
ORDER BY avg_credit DESC
```

Result 5: 'LoanTrendsByAgeGroup.csv'

LoanTrendsByAge...  Open with Numbers		
AGE_GROUP	avg_credit	applicant_count
40-50	6.599946587008505	74401
30-40	6.089564519593123	83117
50 above	6.019148240299853	97187
20-30	4.9218057329799985	52805
0-20	2.97	1

To train the model for deployment I used Amazon SageMaker Studio as it trains a lot of model and with different parameters and helps compare all of them to select the best without use of code. I registered on Amazon SageMaker AutoPilot for single user adhering to the free tier limit. Also, I tried attaching my S3 bucket 's3://bdaminiprojectbucket' to the domain while configuring it. However, I was not able to do it as my bucket was in N. Virginia region and my SageMaker Domain was setup in Stockholm. So, I uploaded the cleaned_application_data.csv file to the bucket that SageMaker automatically creates along with the domain and attached that to the domain.

Key points and takeaways after using SageMaker Studio:

- The Target Column selected for prediction was TARGET, representing a binary classification problem with two categories (0 and 1).
- A total of 10 models were trained, and the performance of each was evaluated against the F1 score, Accuracy, and AUC-ROC metrics.
- The default model achieved an Accuracy of 80.066% and an F1 Score of 23.298%. Other key metrics include:
 - Precision: 17.333%.
 - Recall: 39.098%.
 - AUC-ROC: 0.674.
- Feature importance analysis indicated that:
 - ORGANIZATION_TYPE, AMT_GOODS_PRICE, and DAYS_EMPLOYED were the most influential features in predicting the TARGET column.

Data Bias: The dataset was highly imbalanced as there were more non-debtors (~92%) in the dataset than there were debtors (~8%) . I suspect that this imbalance must have introduced some bias in the models which is the reason why the F1 score is so low even with high accuracy. Looking at the confusion matrix the model predicts True positives effectively as the model had more training data on them.

Here are screenshots of SageMaker Studio at important checkpoints.

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Refresh

More

Select

Build

Analyze

Predict

Deploy

Validating data. Stay on this page until validation is complete. [Learn more](#)

Select a column to predict

Choose the target column. The model that you build predicts values for the column that you select.

Target column

TARGET

Value distribution

Model type

SageMaker Canvas automatically recommends the appropriate model type for your analysis.

2 category prediction

Your model classifies TARGET into two categories.

Configure model

Validating your data...

Validating your data...

application-data-dataset

Random sample: 20.0k rows

Table view

Grid view

Filter

Search

1

Columns

Settings

Refresh

View all

Data visualizer

Column name	Data type	Feature type	Missing	Mismatched
TARGET	123 Numeric	Binary	0.00% (0)	0.00% (0)
REGION_RATING_CLIENT	123 Numeric	-	0.00% (0)	0.00% (0)
ORGANIZATION_TYPE	Text	Categorical	0.00% (0)	0.00% (0)
OCCUPATION_TYPE	Text	Categorical	0.00% (0)	0.00% (0)

Total columns: 19

Total rows: 307,511

Total cells: 5,842,709

Show dropped columns

Model recipe

Add transform

Drop column _c0

AutoML in Analyze Stage:

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Model overview

Your model is being created. Standard build usually takes between 2–4 hours. You can now leave this view.

Time elapsed
44 min 24 sec

Expected build time
45 min

Build type
Standard build

Detailed progress
Training models

application-data-dataset

Total columns: 19

Total rows: 307,511

Total cells: 5,842,709

TARGET

2 category prediction

AutoML Analysis completed:

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Model status

Standard build

Accuracy 80.066%

F1 0.233

The model predicts the correct TARGET 80.066% of the time.

Predict

Deploy

Overview

Scoring

Advanced metrics

Model leaderboard

Column impact

Search columns...

1 ORGANIZATION_TYPE20.118%

2 AMT_GOODS_PRICE16.76%

3 DAYS_EMPLOYED14.193%

4 AMT_CREDIT12.459%

5 AGE_GROUP

Impact on prediction

Impact of ORGANIZATION_TYPE on prediction of TARGET

ORGANIZATION_TYPE

application-data-dataset

Total columns: 19

Total rows: 307,511

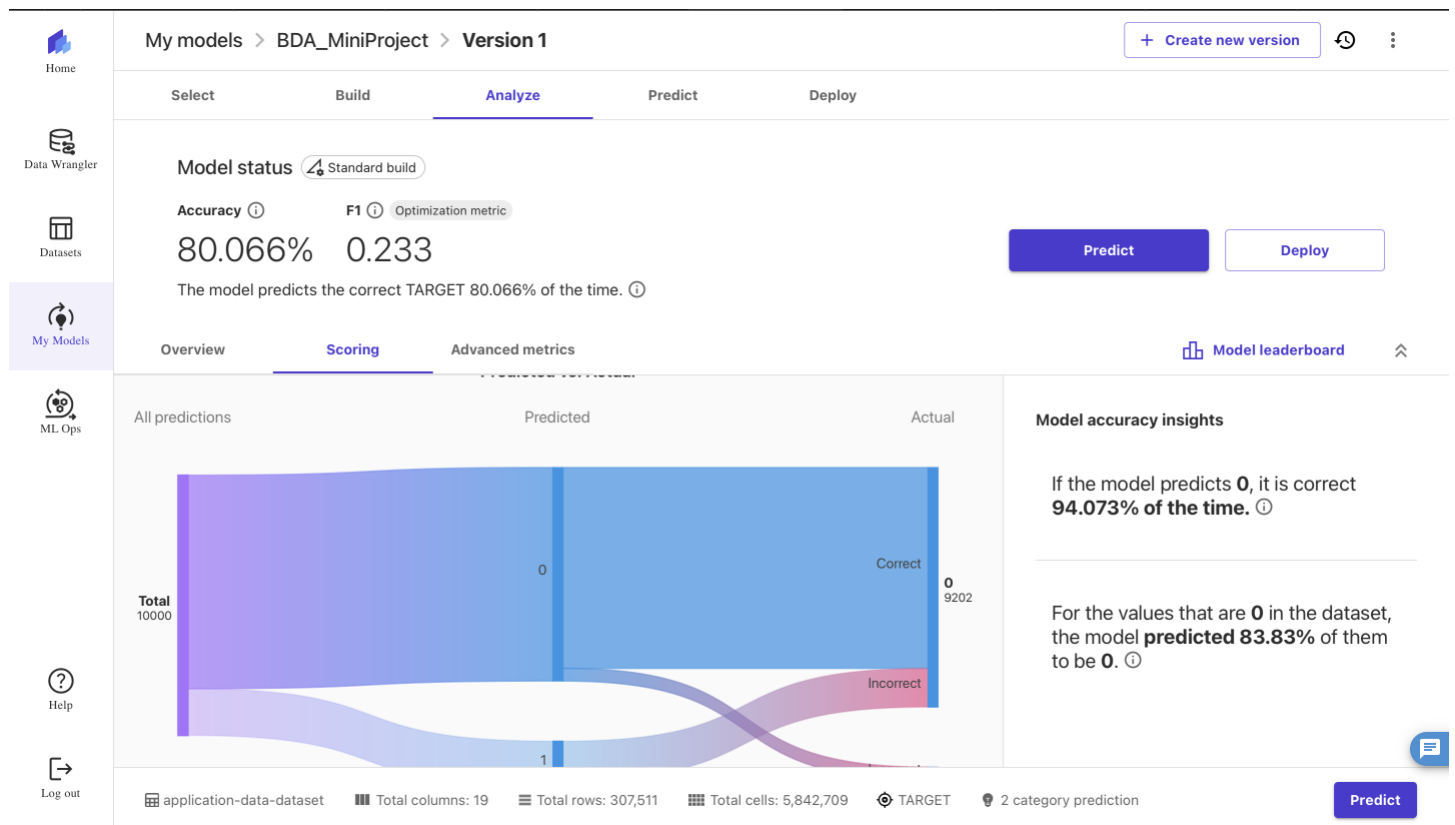
Total cells: 5,842,709

TARGET

2 category prediction

Predict

Models Scoring, this shows us how well the model's predictions are:



The default model had decent accuracy but low F1 score due to high imbalance in the data.

Model Leaderboard:

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My models > BDA_MiniProject > Version 1

+ Create new version

Select

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Model leaderboard

Search leaderboard

Model name	F1	Optimization	Accuracy	AUC	Balanced Accur...	Precision	Recall	Log
FULL-t1296062547469Canvas1734189461067	23.298%	Default model	80.066%	0.676	60.653%	16.898%	37.503%	0.511
FULL-t9296062547469Canvas1734189461067	21.658%		75.757%	0.653	60.138%	14.651%	41.511%	0.54
FULL-t8296062547469Canvas1734189461067	22.990%		77.060%	0.672	61.259%	15.768%	42.417%	0.53
FULL-t7296062547469Canvas1734189461067	22.293%		73.952%	0.668	61.333%	14.683%	46.284%	0.55
FULL-t6296062547469Canvas1734189461067	22.016%		66.055%	0.677	63.000%	13.515%	59.355%	0.62
FULL-t5296062547469Canvas1734189461067	21.977%		66.265%	0.676	62.884%	13.511%	58.852%	0.62
FULL-t4296062547469Canvas1734189461067	22.947%		78.630%	0.674	60.745%	16.184%	39.416%	0.52
FULL-t3296062547469Canvas1734189461067	23.162%		81.154%	0.677	60.188%	17.263%	35.186%	0.50
FULL-t2296062547469Canvas1734189461067	22.293%		73.952%	0.668	61.333%	14.683%	46.284%	
FULL-t10296062547469Canvas1734189461067	22.896%		77.616%	0.671	60.992%	15.857%	41.168%	0.53

Best Performing Model (default):

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+ Create new version

Select

Build

Analyze

Predict

Deploy

Model status

Standard build

Accuracy

80.066%

F1

0.233

Predict

Deploy

The model predicts the correct TARGET 80.066% of the time.

Overview

Scoring

Advanced metrics

Model leaderboard

Positive Class	F1	Optimization metric	Accuracy	Precision	Recall	AUC-ROC
10	24.018%		80.26%	17.333%	39.098%	0.674

Metrics table

Confusion matrix

Precision recall curve

Metrics table

Metric name	Value
precision	0.173
recall	0.391
accuracy	0.803

application-data-dataset

Total columns: 19

Total rows: 307,511

Total cells: 5,842,709

TARGET

2 category prediction

Predict

I used QuickSight for dashboard style visualization. The dashboards built in QuickSight can be easily shared with colleagues using links and are also easy to embed in websites.

After creating account on QuickSight, I integrated by S3 bucket with the cleaned_application_data.csv file with the QuickSight account. After importing the data to SPICE I started data prepping. I built a single sheet informative and insightful dashboard that also presents a good summary of the data used to train the model. I have exported the PDF of the Dashboard as 'QuickSight_Dashboard.pdf' to submit along with this report.

Here is a screenshot of the Data Prep:

The screenshot displays the Amazon QuickSight Data Prep interface. On the left, a sidebar contains a 'Fields' section with a search bar and a list of fields including 'Column-1', 'NAME_HOUSING_TYPE', 'AMT_CREDIT', 'CNT_CHILDREN', 'CNT_FAM_MEMBERS', 'CODE_GENDER', 'NAME_FAMILY_STATUS', 'NAME_EDUCATION_TYPE', 'NAME_INCOME_TYPE', 'REGION_RATING_CLIENT', 'OCCUPATION_TYPE', 'ORGANIZATION_TYPE', 'DAYS_EMPLOYED', 'AMT_GOODS_PRICE', 'AMT_INCOME_TOTAL', and 'AGE_GROUP'. Below this is a 'Community' section with 'Query mode' set to 'SPICE' and '100GB of remaining' storage. The main area shows a 'Data' canvas with a dataset named 'cleaned_application...'. Below the canvas, a 'Dataset' table is visible, showing columns and their data types. The table data includes rows with values for these columns, such as 'House / apa...', '4.066', '0', '1', 'M', 'Single / not ...', 'Secondary / ...', 'Working', '2', and '1'.

Column-1	NAME_HO...	AMT_CREDIT	CNT_CHIL...	CNT_FAM...	CODE_GEN...	NAME_FA...	NAME_ED...	NAME_INC...	REGION_R...	OCCUPATION...
# Integer	< String	Decimal (Floa	# Integer	# Integer	< String	< String	< String	< String	< State	< String
0	House / apa...	4.066	0	1	M	Single / not ...	Secondary / ...	Working	2	L
1	House / apa...	12.935	0	2	F	Married	Higher educ...	State servant	1	C
2	House / apa...	1.35	0	1	M	Single / not ...	Secondary / ...	Working	2	L
3	House / apa...	3.1268	0	2	F	Civil marriage	Secondary / ...	Working	2	L
4	House / apa...	5.13	0	1	M	Single / not ...	Secondary / ...	Working	2	C
5	House / apa...	4.905	0	2	M	Married	Secondary / ...	State servant	2	L
6	House / ana...	15.6073	1	3	F	Married	Higher educ...	Commercial	2	L

I excluded the 'column-1' from the dataset and readjusted the datatype for some of the columns.

Dashboard Walkthrough:

The Dashboard aims to provide a clear insightful understanding of the dataset. Specifically, the factors influencing an individual defaulting on the loan. Every visualization has the corresponding tooltip picture below it as well.

Bank Loan Default Analysis

Big Data Applications (Fall 2024)
Author: Rishikesh Kakde (rkakde)

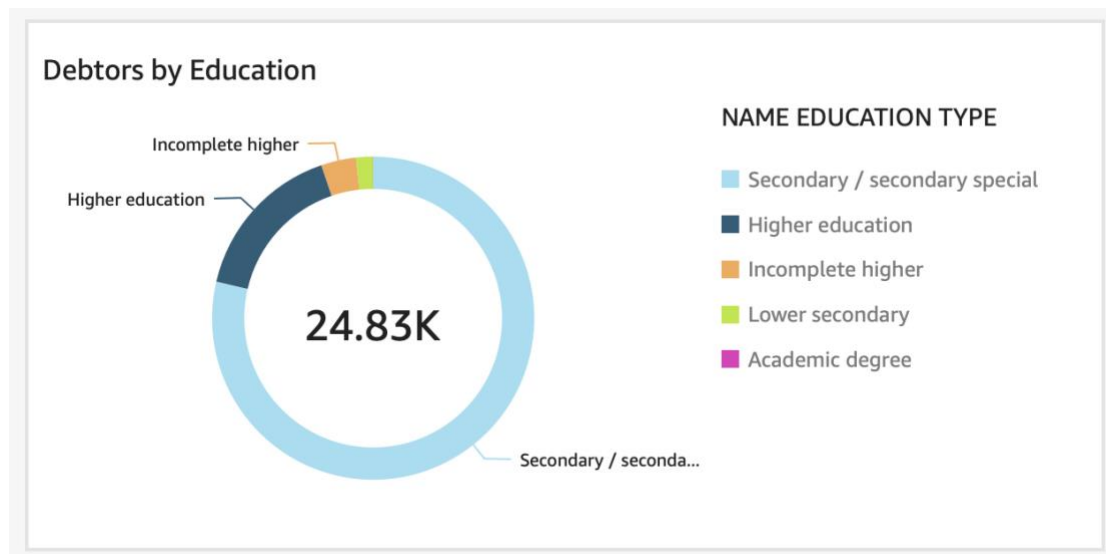
Target 1: Debtors (Loan Default)
Target 0: Non-Debtors

Debtors %

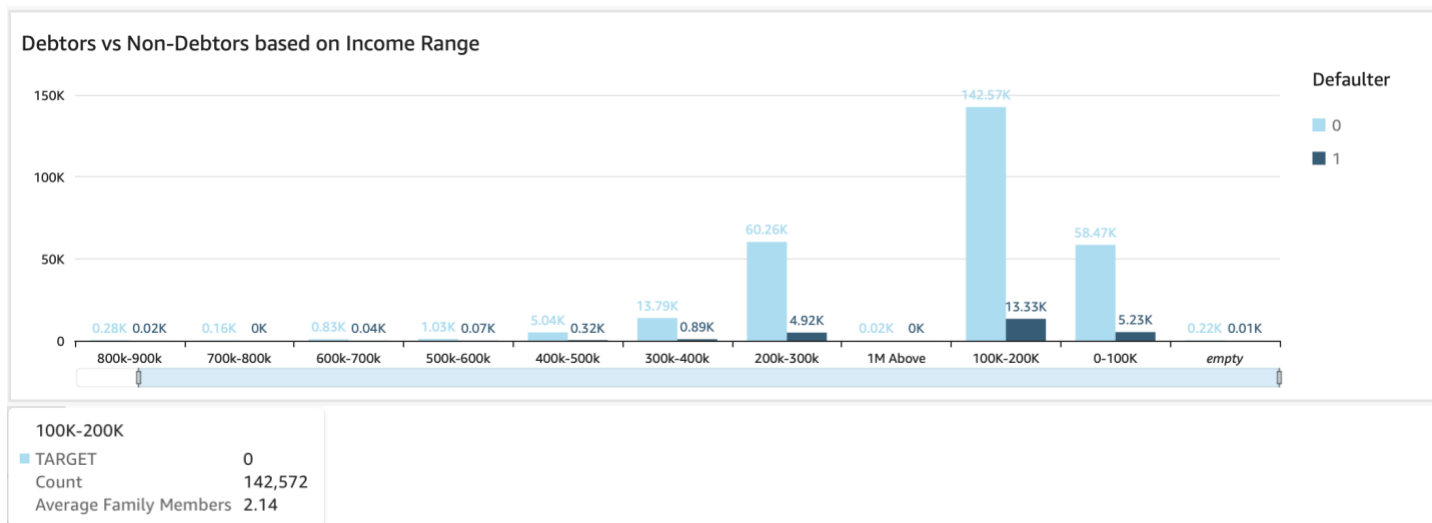
8.07%

This is the header of the Dashboard. It contains the title of the project and author information. It also has a KPI on the right end signifying that the data had 8.07% individuals who defaulted on the loan. That gives a very high level summary of the data.

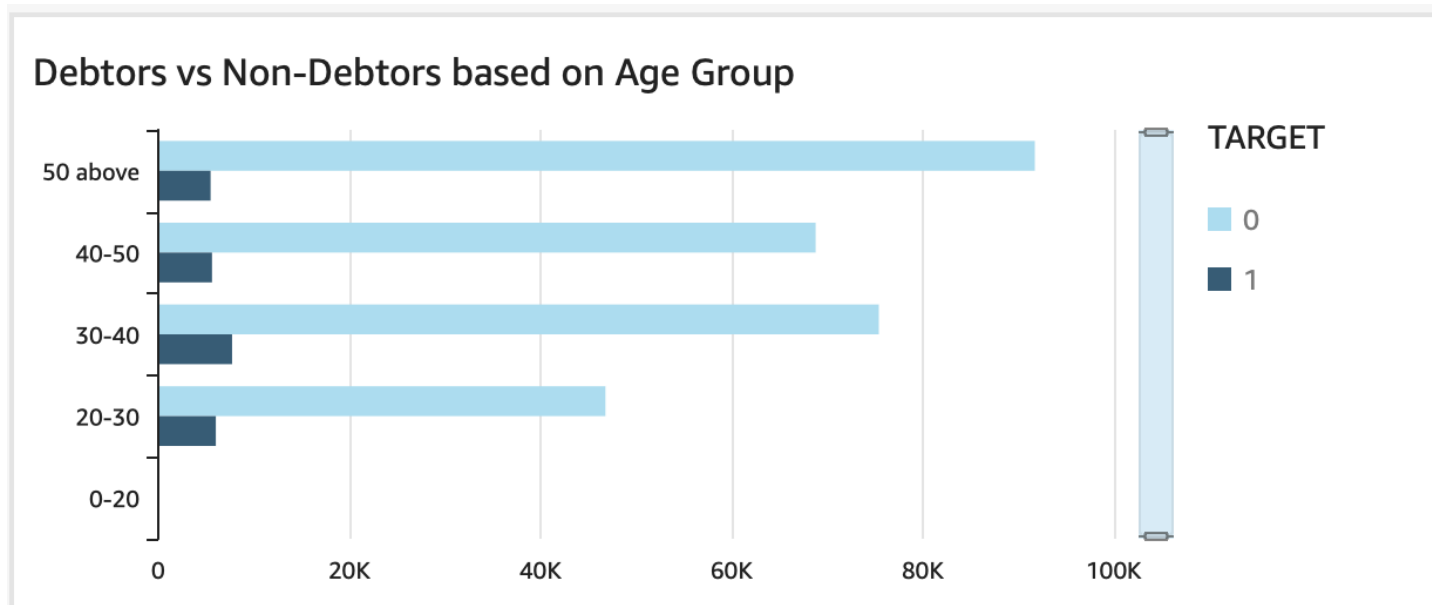
Note that sheet-wide filter for age group, income bracket and children count was provided as they were found to be important in predicting the probability of an individual paying back the loan.



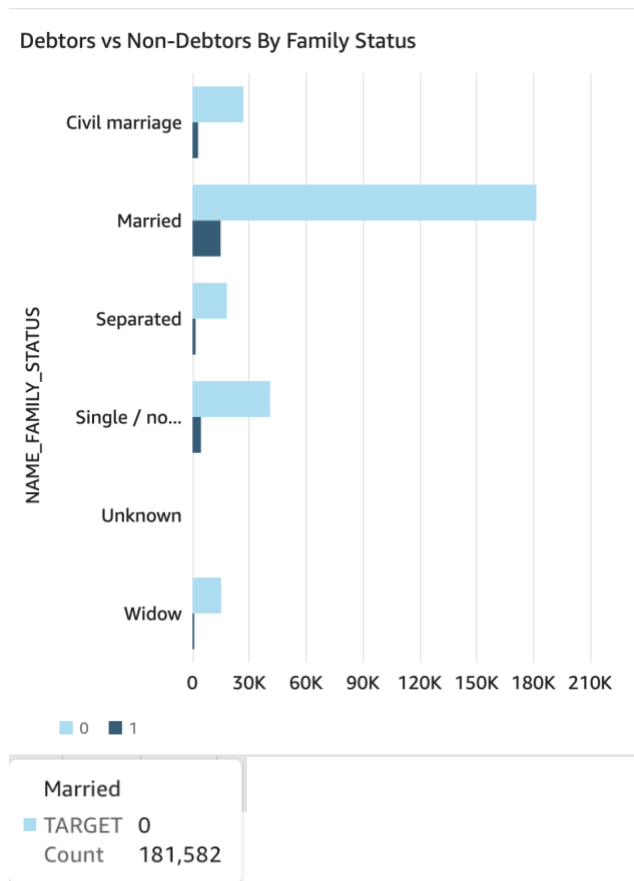
This donut chart highlights the type of education of individual debtors. The tool tip shows the percentage of total debtors with that education type. This helps us understand that, based on this data, most of the individuals who default on their loan had secondary or secondary special education. Individuals with lower secondary education who defaulted on their loan were the smallest part in this dataset.



This bar chart above plots the loan amount vs income bracket and groups them by debtor status. As specified in the title card, one (1) represents individuals who defaulted on their loan and zero (0) represents individuals who didn't. This plot tells us that most of the loans are secured by people within the income range 100k-200k. The highest number of people defaulting on their loan also come from this category which is a very small percentage compared to the people who do pay it back. People with very high or very low income are not even close to this income range in terms of borrowing money. The tooltip shows the actual count for the highlighted bar and also displays the average family members count for that category.



This next visualization, which is a bar chart, plots age group vs amount borrowed grouped by amount borrowed. In this the number of people defaulting on the loan is consistent across all age groups. However, we see that the individuals with age 50 or above have a higher percentage of successfully paying back the loan compared to all the other age categories.



The next visualization, which is a horizontal bar chart, plots family status vs amount borrowed grouped by debtor status. This plot highlights that the married people borrow the most amount of money, at the same time, they default the most amount of time too. Other categories do not even come close to married category in terms of amount borrowed.

Results:

The rigorous analysis helped us understand that the most common factors that influence the ability of an individual to successfully pay off a loan are type of organization the individual is employed at, amount of money borrowed, the length of employment, age group, and education.

SageMaker Studio quickly built the models and gave a good overview of application of machine learning on the prediction task. The imbalance in the data between debtors and non-debtors proved to be a challenge in training the model. SageMaker

Studio does test good amount of models but it takes too long to train them. Python implementation of the models will give much more control to the designer and for this task meticulous selection of hyperparameters and model would have been faster.

The QuickSight visualizations helped in understanding the high impact of age group, family status in prediction. The visualizations highlighted that most of the individuals who default on their loan had secondary or secondary special education, individuals with age 50 or above have a higher percentage of successfully paying back the loan, and married people borrow the most amount of money, at the same time, they default the most amount of time too.

Conclusion:

This project successfully demonstrated the end-to-end process of building a scalable and insightful big data analytics pipeline. Starting with raw loan application data from Kaggle, we utilized Amazon Web Services (AWS) and Apache PySpark to process, clean, and transform the dataset into a format suitable for analysis and modeling. The project also explored machine learning capabilities using AWS SageMaker Autopilot and visualized key insights using AWS QuickSight.

The analysis revealed critical factors influencing loan default, including the type of organization, amount borrowed, employment length, age group, and education level. SageMaker's AutoML capabilities enabled us to achieve a high accuracy of 80%, but the low F1 score highlighted the need for better handling of class imbalances.

The QuickSight dashboard provided insights into borrower demographics and behaviors, highlighting trends such as higher loan defaults among those with secondary education and higher success rates for individuals aged 50 and above. Despite some limitations with SageMaker's AutoML, the project achieved its objectives of building a model and deriving insights on loan defaults.

References:

1. Dataset Source: [Kaggle](#)
2. Apache PySpark Documentation: [PySpark Official Documentation](#)
3. AWS S3: [Link to AWS official guide](#)
4. AWS SageMaker Autopilot: [SageMaker Autopilot Documentation](#)
5. AWS QuickSight: [QuickSight Documentation](#)
6. Apache Hadoop AWS Integration: [Hadoop AWS Documentation](#)